THE COLOURED SAND THEOREM: THE p-PART OF THE SANDPILE GROUP OF THE HAMMING GRAPH

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ABSTRACT. Let $H(d,q) = K_q^d$ be the Hamming graph and $K(H(d,q))$ its sandpile group (critical group). For primes $p \nmid q$ its p-part was determined by Jacobson, Niedermaier and Reiner in 2003, who called the remaining primes difficult, "due to bad behavior of its p-primary structure". We treat the primes $p \mid q$. Jacobson, Niedermaier and Reiner showed that over $\mathbb{Z}$ the Laplacian splits into explicit triangular pieces; we read these pieces as operators $q(c+H)+U$ on the Boolean lattice (A). From this we prove: (B) a closed formula for the Sylow p-subgroup for every prime $p > d$ and every q divisible by p, in which ballot numbers appear; (C) the exponent of the Sylow p-subgroup for every prime p and every $q \geq 3$, namely $p^{\max_{1\leq s\leq d}(es+v_p(s))}$ with $e = v_p(q) \geq 1$; (D) the number of its cyclic factors when $p \mid q$, from Wilson's diagonal form of inclusion matrices. For $d \leq 3p-1$ (p odd) we determine the group exactly (F, G), by writing down the lifts to the p-adic integers of the tilting modules of SL_2 in characteristic p. This proves the Glue Conjecture of version 1, which had predicted 44 cases before they were computed, all correctly. We conjecture that the same formula holds for $d \leq p^2-2$ (H, p odd), and we give a closed form for the number of cyclic factors when $3 \mid q$.

1. INTRODUCTION

For a connected graph Γ with Laplacian $L(\Gamma)$, the sandpile group $K(\Gamma)$ is the torsion subgroup of $\operatorname{coker} L(\Gamma)$, and $\operatorname{coker} L(\Gamma) \cong \mathbb{Z} \oplus K(\Gamma)$; its order is the number of spanning trees [Big99]. The Hamming graph $H(d,q)$ has vertex set $(\mathbb{Z}/q)^d$, two vertices being adjacent when they differ in exactly one coordinate; it is the Cartesian power K_q^d of the complete graph. Its Laplacian eigenvalues are qs, $0 \leq s \leq d$, with multiplicity $\binom{d}{s}(q-1)^s$.

Jacobson, Niedermaier and Reiner [JNR03] determined the Sylow p-subgroup of $K(K_{n_1} \times \cdots \times K_{n_k})$ for every prime p that divides none of the n_i [JNR03, Theorem 2], and bounded the number of invariant factors [JNR03, Theorem 3]. About the other primes they wrote: "Describing the exact structure of the critical group $K(K_{n_1} \times \cdots \times K_{n_k})$ appears to be difficult, due to bad behavior of its p-primary structure for primes p that divide some of $n_1, \ldots, n_k$." Ducey and Jalil [DJ14] recovered the results of [JNR03] by another method. For $q = 2$ the graph is the hypercube; its odd part is due to Bai [Bai03] and its 2-part was determined in [DST].

In this paper $q \geq 2$ is arbitrary, p is a prime dividing q, and $e := v_p(q) \geq 1$, except where we say otherwise.

The pieces. Let B_m be the free abelian group on the subsets of $[m] = \{1, \ldots, m\}$, graded by size: $B_m = \bigoplus_{s=0}^{m} B_m(s)$. Let H act on $B_m(s)$ as the scalar s, and let U and D be the up and down operators,

$$U(S) = \sum_{x \notin S} S \cup \{x\}, \qquad D(S) = \sum_{x \in S} S \setminus \{x\}.$$

For integers $c \geq 0$ put

$$X_q(m,c) := \operatorname{coker}\big(q(c+H)+U \ \text{ on } B_m\big).$$

It is finite when $c \geq 1$; for $c = 0$ it is $\mathbb{Z}$ plus a finite group.

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Theorem A (Jacobson–Niedermaier–Reiner [JNR03, Theorem 13])**.** *For every $q \geq 2$ and $d \geq 1$, as abelian groups,*

$$\operatorname{coker} L\big(H(d,q)\big) \;\cong\; \bigoplus_{j=0}^{d} X_q(d-j,\, j)^{\oplus \binom{d}{j}(q-2)^j}.$$

This is the case $n_1 = \cdots = n_d = q$ of [JNR03, Theorem 13]: there the pieces are upper triangular matrices $M(N; n_S)$ of size $2^{|S|}$ with identity blocks above the diagonal, and after reversing the order of the subsets of S they are the matrices $q(c + H) + U$. We include the short proof below, in the form we use. For $q = 2$ only $j = 0$ survives and the theorem says $\operatorname{coker} L(Q_d) \cong X_2(d, 0)$, the starting point of [DST].

Theorem B (the primes $p > d$)**.** *Let $d \geq 1$, let $p > d$ be a prime dividing q, and $e = v_p(q)$. Then*

$$\operatorname{Syl}_p K\big(H(d,q)\big) \;\cong\; \bigoplus_{v=1}^{d} \big(\mathbb{Z}/p^{ev}\big)^{n_v}, \qquad n_v = \sum_{\substack{j+2k=d+1-v \\ (j,k)\neq(0,0)}} \binom{d}{j} (q-2)^j\, b(k,v),$$

the sum over integers $j, k \geq 0$, where $b(k,v) = \frac{v}{v+2k}\binom{v+2k}{k} = [x^k]\, C(x)^v$ are the ballot numbers and $C(x) = \sum_n \frac{1}{n+1}\binom{2n}{n} x^n$ is the Catalan series.

Equivalently, with $c(z) = C(z^2)$ and $u = q - 2$, $\sum_v n_v y^v = [z^{d+1}]\,(1+uz)^d\, \frac{yz\,c(z)}{1-yz\,c(z)} - y^{d+1}$. For instance, with $u = q - 2$:

$$\begin{aligned}
d = 2: &\quad (\mathbb{Z}/p^e)^{u^2+1} \oplus (\mathbb{Z}/p^{2e})^{2u},\\
d = 3: &\quad (\mathbb{Z}/p^e)^{u^3+3u} \oplus (\mathbb{Z}/p^{2e})^{3u^2+2} \oplus (\mathbb{Z}/p^{3e})^{3u},\\
d = 4: &\quad (\mathbb{Z}/p^e)^{u^4+6u^2+2} \oplus (\mathbb{Z}/p^{2e})^{4u^3+8u} \oplus (\mathbb{Z}/p^{3e})^{6u^2+3} \oplus (\mathbb{Z}/p^{4e})^{4u}.
\end{aligned}$$

Theorem B is the case $m < p$ of a statement about single pieces (Theorem E in Section 3), which also gives, for every prime p, the contribution of all pieces with $d - j < p$.

Theorem C (the exponent)**.** *Let $q \geq 3$, let p be any prime and $e = v_p(q)$. The exponent of $\operatorname{Syl}_p K(H(d,q))$ is p^E, where*

$$E = \max_{1\leq s\leq d} \big(e\,s + v_p(s)\big) \quad \text{if } e \geq 1, \qquad E = \lfloor \log_p d \rfloor \quad \text{if } e = 0.$$

For $p > d$ this gives $E = ed$, as in Theorem B. For $p \leq d$ the maximum need not be at $s = d$: for $q = 3$ and $d = 10$ it is at $s = 9$ and $E = 11$. The case $e = 0$ ($p \nmid q$) can also be read off from [JNR03, Theorem 2]; we include it because the proof below covers it at no cost.

Theorem D (the number of cyclic factors)**.** *Let $p \mid q$. The number of cyclic factors of $\operatorname{Syl}_p K(H(d,q))$ is*

$$N = \sum_{j=0}^{d} \binom{d}{j} (q-2)^j \Big(2^{\,d-j} - \rho_p(d-j)\Big) \;-\; 1, \qquad \rho_p(m) = \sum_{s=0}^{m-1} r_p(m,s),$$

where $r_p(m,s) = \sum_i \big(\binom{m}{i} - \binom{m}{i-1}\big)$, the sum over $0 \leq i \leq t$ with $p \nmid k - i$, and $(t,k) = (s, s+1)$ if $2s \leq m-1$, $(t,k) = (m-1-s, m-s)$ otherwise.

Here $\rho_p(m)$ is the rank over $\mathbb{F}_p$ of U on B_m. For $q = p = 3$ the theorem gives $N = 1, 4, 12, 34, 96, 274, \ldots$, and $N = 3^{d-1} + 2^{d-1} - 1$ for every d (Corollary 6). For $q = p$ the value of N equals the bound $\frac{1}{2}(q^d + (q-2)^d) - 1$ of [JNR03, Theorem 3] for $d \leq 2$, is one less for $d = 3$, and is strictly smaller for $4 \leq d \leq 8$ and $p \leq 11$ (where we compared).

Corollary 1. *$\operatorname{Syl}_p K(H(d,q))$ depends on q only through $e = v_p(q)$ and the multiplicities $\binom{d}{j}(q-2)^j$.*

Status. This is a preprint. The proofs of Theorems B–D and E were re-read once by the system that wrote them (see Provenance) and checked against the computations of Section 7; they have not yet been read by an independent reader. Version 2 (13 October 2026) adds Theorems F and G, Conjecture H and Corollary 6; in version 1, Theorem F was stated as a conjecture. The new proofs have been read only by the system that wrote them. Conjecture H is supported by the method of proof and by computation; see Section 8.

By Theorem G (Section 6), what remains is the range $d \geq 3p$, and there only the pieces $X_q(m,c)$ with $m \geq 3p$. For $m \leq p^2-2$ we state a precise conjecture; beyond that range the tilting modules have four or more Weyl factors, and we leave the question open.

2. The splitting (after Jacobson, Niedermaier and Reiner)

Proof of Theorem A. Let $K_q = qI - J$ be the Laplacian of the complete graph K_q, acting on $\mathbb{Z}^q$ with basis $e_0,\dots,e_{q-1}$, and put $f_i = e_i - e_0$ $(1 \leq i \leq q-1)$ and $\sigma = f_1 + \cdots + f_{q-1}$. Then $e_0, \sigma, f_1, \dots, f_{q-2}$ is a basis of $\mathbb{Z}^q$ (unimodular change from $e_0, f_1, \dots, f_{q-1}$), and

$$K_q e_0 = -\sigma, \qquad K_q \sigma = q\sigma, \qquad K_q f_i = q f_i.$$

So $\mathbb{Z}^q = A \oplus W$ with $A = \mathbb{Z}e_0 \oplus \mathbb{Z}\sigma$ and $W = \bigoplus_{i \leq q-2} \mathbb{Z} f_i$, both stable under K_q, and K_q is the scalar q on W. The Laplacian of $H(d,q)$ is $L = \sum_{i=1}^{d} 1 \otimes \cdots \otimes K_q \otimes \cdots \otimes 1$ on $(\mathbb{Z}^q)^{\otimes d} = \bigoplus_{T \subseteq [d]} A^{\otimes([d]\setminus T)} \otimes W^{\otimes T}$, and each summand is L-stable. On the summand of T with $|T| = j$, L acts as $(L_A + qj) \otimes 1$, where $L_A = \sum_i K_q^{(i)}$ on $A^{\otimes(d-j)}$ and $W^{\otimes T} \cong \mathbb{Z}^{(q-2)^j}$. There are $\binom{d}{j}$ sets T of size j.

Index the basis of $A^{\otimes m}$ by the subsets $S \subseteq [m]$, the factor in position i being σ if $i \in S$ and e_0 otherwise. Then $L_A(S) = q|S|\, S - \sum_{x \notin S} S \cup \{x\}$, that is, $L_A = qH - U$, and $L_A + qj = q(j+H) - U$. Conjugating by the diagonal matrix $(-1)^{|S|}$ turns $-U$ into U. Hence the summand contributes $\operatorname{coker}(q(j+H) + U \text{ on } B_{d-j}) = X_q(d-j,j)$, $\binom{d}{j}(q-2)^j$ times. □

Proof of Corollary 1. By Theorem A it suffices to see that $\operatorname{Syl}_p X_q(m,c)$ depends only on the p-adic valuations of the diagonal entries $a_s = q(c+s)$. This is Lemma 2. □

Lemma 2 (only the valuations matter)**.** *Let $a_0,\dots,a_m$ be elements of $\mathbb{Z}_{(p)}$ and $M = \operatorname{diag}(a_{|S|}) + U$ on $B_m \otimes \mathbb{Z}_{(p)}$. If $a_s = p^{w_s} u_s$ with units u_s (or $a_s = 0$, with $u_s = 1$), then M is equivalent over $\mathbb{Z}_{(p)}$ to*

$$\operatorname{diag}(p^{w_{|S|}}) + U \qquad (\textit{with } p^w := 0 \textit{ when } a_s = 0).$$

Proof. Look for units λ_s, μ_s such that $P = \operatorname{diag}(\lambda_{|S|})$ and $Q = \operatorname{diag}(\mu_{|S|})$ give $PMQ = \operatorname{diag}(p^{w_{|S|}}) + U$. The diagonal of PMQ on rank s is $\lambda_s a_s \mu_s$ and its block from rank s to rank $s+1$ is $\lambda_{s+1}\mu_s U$, so we need $\lambda_s \mu_s u_s = 1$ and $\lambda_{s+1}\mu_s = 1$. Put $\mu_0 = 1$, $\lambda_0 = u_0^{-1}$, and recursively $\lambda_{s+1} = \mu_s^{-1}$, $\mu_{s+1} = (\lambda_{s+1} u_{s+1})^{-1}$. □

3. Strings, and the primes $p > m$

On B_m one has $[D,U] = m - 2H$: on a set S of size s, $DU(S) = (m-s)S + \Sigma$ and $UD(S) = s\,S + \Sigma$ with the same sum $\Sigma = \sum_{y \in S,\, x \notin S} (S \setminus \{y\}) \cup \{x\}$. For $0 \leq k \leq m/2$ let $\Lambda_k = \ker D \cap B_m(k)$. By induction on i, for $v \in \Lambda_k$,

$$D\,U^i v = i\,(m - 2k - i + 1)\,U^{i-1} v. \tag{1}$$

Over $\mathbb{Q}$ this is the classical $\mathfrak{sl}_2$-structure of the Boolean lattice [Pro82]:

$$B_m \otimes \mathbb{Q} = \bigoplus_{k \leq m/2} \bigoplus_{i=0}^{m-2k} U^i \Lambda_k^{\mathbb{Q}},$$

$\dim \Lambda_k^{\mathbb{Q}} = \binom{m}{k} - \binom{m}{k-1}$, and $U^{m-2k+1}\Lambda_k = 0$. We call $\Lambda_k, U\Lambda_k, \dots, U^{m-2k}\Lambda_k$ a *string*; its ranks are $k,\dots,m-k$ and its length is $m-2k+1$.

Lemma 3 (strings over $\mathbb{Z}_{(p)}$)**.** *If $p > m$, then $B_m \otimes \mathbb{Z}_{(p)} = \bigoplus_{k\le m/2} \bigoplus_{i=0}^{m-2k} U^i\Lambda_k$, each Λ_k is free of rank $\binom{m}{k} - \binom{m}{k-1}$, and $U^i\colon \Lambda_k \to U^i\Lambda_k$ is an isomorphism for $i \le m-2k$.*

Proof. All lattices below are over $R = \mathbb{Z}_{(p)}$. First, for $r \le m/2$ we show $B_m(r) = UB_m(r-1) \oplus \Lambda_r$, by induction on r (for $r = 0$, $\Lambda_0 = B_m(0)$). By induction $B_m(r-1) = \bigoplus_{k<r} U^{r-1-k}\Lambda_k$, and by (1) the operator DU acts on $U^{r-1-k}\Lambda_k$ as the scalar $(r-k)(m-k-r+1)$; both factors lie in $[1, m]$, so they are units. Hence DU is invertible on $B_m(r-1)$, $\pi = U(DU)^{-1}D$ is an idempotent on $B_m(r)$ with image $UB_m(r-1)$, and its kernel is $\ker D \cap B_m(r) = \Lambda_r$ because $U(DU)^{-1}$ is injective. This gives the decomposition, and Λ_r is a direct summand, so free, of rank $\binom{m}{r} - \binom{m}{r-1}$.

Now fix any rank r and consider $\Phi\colon \bigoplus_{k\le \min(r,m-r)} \Lambda_k \to B_m(r)$, $(v_k) \mapsto \sum_k U^{r-k}v_k$. For $k' \le r$ let $\psi_{k'}\colon B_m(r) \to \Lambda_{k'}$ be $D^{r-k'}$ followed by the projection of $B_m(k') = UB_m(k'-1) \oplus \Lambda_{k'}$ onto $\Lambda_{k'}$. By (1), $D^{r-k'}U^{r-k}v_k$ is 0 if $k > k'$ (since $Dv_k = 0$), lies in $U^{k'-k}\Lambda_k \subseteq UB_m(k'-1)$ if $k < k'$, and equals $\prod_{i=1}^{r-k} i(m-2k-i+1)\, v_k$ if $k = k'$, a unit multiple of v_k (all factors lie in $[1, m]$ because $k \le m-r$). So $(\psi_{k'}) \circ \Phi$ is triangular with unit diagonal, Φ is split injective, and its image is a direct summand of $B_m(r)$ of full rank (compare ranks over $\mathbb{Q}$). Hence Φ is an isomorphism. Finally $U^{m-2k+1}\Lambda_k = 0$ holds over $\mathbb{Q}$, hence over R. □

Each string is stable under $q(c+H)+U$: in the basis $v, Uv, \dots, U^{\ell-1}v$ ($\ell = m-2k+1$) the operator is lower bidiagonal with diagonal entries $q(c+k+i)$, $0 \le i < \ell$, and all subdiagonal entries 1. The cokernel of such a matrix over a principal ideal domain is cyclic of order its determinant, because the minor obtained by deleting the first row and last column equals 1; if one diagonal entry is 0 (only possible for $c = k = i = 0$) the cokernel is $\mathbb{Z}_{(p)}$. This proves:

Theorem E (pieces with $m < p$)**.** *Let p be any prime, $e = v_p(q)$, and $m < p$. Then*

$$\operatorname{Syl}_p X_q(m,c) \;\cong \bigoplus_{\substack{0\le k\le m/2\\ (c,k)\neq(0,0)}} \big(\mathbb{Z}/p^{A_k}\big)^{\binom{m}{k}-\binom{m}{k-1}}, \qquad A_k = \sum_{r=k}^{m-k} \big(e + v_p(c+r)\big),$$

and for $c = 0$ the string $k = 0$ contributes only the free summand $\mathbb{Z}$.

Proof of Theorem B. If $p > d$, every piece $X_q(d-j, j)$ has $m = d-j < p$ and $c+r \le d < p$, so $v_p(c+r) = 0$ for $c+r \ge 1$ and $A_k = e(m-2k+1)$. Put $v = m-2k+1 = d+1-j-2k$; then $\binom{m}{k} - \binom{m}{k-1} = \binom{v+2k-1}{k} - \binom{v+2k-1}{k-1} = \frac{v}{v+2k}\binom{v+2k}{k} = b(k,v)$. Summing over j with Theorem A gives the formula. The generating function follows from $\sum_k b(k,v)x^k = C(x)^v$. □

4. The exponent

Lemma 4. *Let $c \ge 1$ and $M = q(c+H)+U$ on B_m. Then M is invertible over $\mathbb{Q}$, and for $S \subseteq T$ with $|S| = s$, $|T| = s+k$,*

$$(M^{-1})_{T,S} = (-1)^k\, k! \prod_{i=0}^{k} \frac{1}{q(c+s+i)},$$

while $(M^{-1})_{T,S} = 0$ if $S \not\subseteq T$.

Proof. With $\Delta = q(c+H)$, $M^{-1} = (1+\Delta^{-1}U)^{-1}\Delta^{-1} = \sum_{k\ge0}(-\Delta^{-1}U)^k\Delta^{-1}$, a finite sum. A path from S to T in $(\Delta^{-1}U)^k\Delta^{-1}$ is a maximal chain $S = S_0 \subset \cdots \subset S_k = T$, there are $k!$ of them, and each collects Δ^{-1} once at every rank from s to $s+k$. □

For a nonsingular square matrix over a discrete valuation ring with uniformizer p, the exponent of the cokernel is p^ϵ with $-\epsilon$ the least valuation of an entry of M^{-1} (write M in Smith form). Hence, for $c \ge 1$, the exponent of $\operatorname{Syl}_p X_q(m,c)$ is p to the power

$$\max_{\substack{s,k\ge0\\ s+k\le m}} \Big(\sum_{i=0}^{k} \big(e + v_p(c+s+i)\big) - v_p(k!)\Big) = \max_{c\le a\le b\le c+m} \Big(eL + v_p(L) + v_p\tbinom{b}{L}\Big), \quad L = b-a+1,$$

where $a = c + s$, $b = c + s + k$, and we used $\frac{b!}{(a-1)!\,(b-a)!} = L\binom{b}{L}$.

Lemma 5 (carries)**.** *For $d \geq 1$ and $e \geq 1$,*

$$\max_{1 \leq L \leq b \leq d} \Big(eL + v_p(L) + v_p\tbinom{b}{L} \Big) = \max_{1 \leq s \leq d} \big(es + v_p(s) \big),$$

and for $e = 0$ the left side equals $\lfloor \log_p d \rfloor$.

Proof. Taking $L = b = s$ gives $\geq$. Let $\lambda = \lfloor \log_p b \rfloor$. By Kummer's theorem $v_p\binom{b}{L}$ is the number of carries in the addition $L + (b - L)$ in base p. No carry occurs at a position below $v_p(L)$, where the digits of L vanish, and a carry out of position i forces $b \geq p^{i+1}$, so $i \leq \lambda - 1$. Hence $v_p(L) + v_p\binom{b}{L} \leq \lambda$. For $e = 0$ this gives the bound $\lfloor \log_p d \rfloor$, attained at $L = b = p^{\lfloor \log_p d \rfloor}$.

Let $e \geq 1$; we induct on b, the case $b = 1$ being clear. Let $s^* = p^\lambda \lfloor b/p^\lambda \rfloor$, so $1 \leq s^* \leq b < s^* + p^\lambda$ and $v_p(s^*) \geq \lambda$. If $L \leq s^*$, then $eL + v_p(L) + v_p\binom{b}{L} \leq es^* + \lambda \leq es^* + v_p(s^*)$. If $L > s^*$, write $b = s^* + r$ and $L = s^* + r'$ with $1 \leq r' \leq r < p^\lambda$. Then $v_p(L) = v_p(r')$, and since $(b - L) + r' = r < p^\lambda$, the additions $L + (b - L)$ and $r' + (r - r')$ have the same carries, so $v_p\binom{b}{L} = v_p\binom{r}{r'}$. By induction there is $1 \leq t \leq r$ with $er' + v_p(r') + v_p\binom{r}{r'} \leq et + v_p(t)$. Since $t < p^\lambda \leq p^{v_p(s^*)}$, $v_p(s^* + t) = v_p(t)$, and so $eL + v_p(L) + v_p\binom{b}{L} \leq e(s^* + t) + v_p(s^* + t)$ with $s^* + t \leq b$. □

Proof of Theorem C. By Theorem A the exponent is the largest exponent of the pieces $\mathrm{Syl}_p X_q(d - j, j)$ that occur. For $j \geq 1$ the piece occurs ($q \geq 3$) and the computation above gives the maximum over $j \leq a \leq b \leq d$; for $j = 1$ this is the full range $1 \leq a \leq b \leq d$. For $j = 0$, the row of the empty set in $M_0 = qH + U$ is zero and M_0 maps into $B' = \bigoplus_{s \geq 1} B_d(s)$, so

$$\operatorname{coker} M_0 = \mathbb{Z} \oplus B'/(M'' B' + \mathbb{Z}u),$$

where M'' is the restriction of M_0 to B' (nonsingular, lower triangular by rank with diagonal qs) and $u = M_0(\emptyset)$. Its torsion is a quotient of $\operatorname{coker} M''$, whose exponent is given by Lemma 4 (with $c = 0$ and $s \geq 1$) as the same maximum over $1 \leq a \leq b \leq d$. So the exponent is p to $\max_{1 \leq a \leq b \leq d}(eL + v_p(L) + v_p\binom{b}{L})$, and Lemma 5 finishes. □

For $q = 2$ the same argument gives only an upper bound, and it is not attained: for Q_2 the group is $\mathbb{Z}/4$ while the bound is 2^3.

5. The number of cyclic factors

Proof of Theorem D. The number of cyclic factors of $\mathrm{Syl}_p K$ is $\dim_{\mathbb{F}_p}(\operatorname{coker} L \otimes \mathbb{F}_p) - 1$, the 1 for the free summand. Modulo p, $q(j + H) + U \equiv U$, so by Theorem A $\dim_{\mathbb{F}_p}(\operatorname{coker} L \otimes \mathbb{F}_p) = \sum_j \binom{d}{j}(q-2)^j \big(2^{d-j} - \operatorname{rank}_{\mathbb{F}_p} U|_{B_{d-j}}\big)$. The rank of U is the sum over s of the ranks of the inclusion matrices $W_{s,s+1}$ of s-subsets versus $(s+1)$-subsets of $[m]$. By Wilson's diagonal form [Wil90], for $t \leq \min(k, m-k)$ the matrix $W_{t,k}$ is equivalent over $\mathbb{Z}$ to a diagonal matrix with entries $\binom{k-i}{t-i}$ of multiplicity $\binom{m}{i} - \binom{m}{i-1}$, $0 \leq i \leq t$ (we use the statement as given in [CSX15, (2)]); so its rank over $\mathbb{F}_p$ is $\sum_{i:\, p \nmid \binom{k-i}{t-i}} \big(\binom{m}{i} - \binom{m}{i-1}\big)$. For $W_{s,s+1}$ with $2s \leq m - 1$ take $(t, k) = (s, s+1)$, so $\binom{k-i}{t-i} = s + 1 - i$; otherwise complementation identifies $W_{s,s+1}$ with the transpose of $W_{m-s-1,m-s}$ and we take $(t, k) = (m - 1 - s, m - s)$. This is $r_p(m, s)$. □

Corollary 6 (the number of factors when $3 \mid q$)**.** *If $3 \mid q$, the number of cyclic factors of $\mathrm{Syl}_3 K(H(d, q))$ is*

$$N = (q-1)^d + \frac{q^d - (q-1)^d}{3} - \frac{(q-1)^d - (q-3)^d}{6} - 1.$$

In particular $N = 3^{d-1} + 2^{d-1} - 1$ for $q = 3$.

Proof. By Theorem D it suffices to know $f(m) = \dim_{\mathbb{F}_3} \operatorname{coker}(U|_{B_m \otimes \mathbb{F}_3})$. Identify $B_m \otimes \mathbb{F}_3$ with $V^{\otimes m}$, $V = \mathbb{F}_3^2$, so that $U = \sum_i e_i$ with e the nilpotent 2×2 Jordan block acting on the i-th factor. The e_i commute, so $U^3 = \sum_i e_i^3 = 0$, and $V^{\otimes m}$ is a module over the Hopf algebra $\mathbb{F}_3[e]/(e^3)$ with e primitive. Then $f(m)$ is the number of Jordan blocks of U. Write J_a for the indecomposable module of dimension $a \leq 3$. Then:

- $J_1 \otimes J_2 = J_2$;
- $J_2 \otimes J_2 = J_3 \oplus J_1$, since $e^2(v_0 \otimes v_0) = 2\, v_1 \otimes v_1 \neq 0$;
- $J_3 \otimes J_2 = J_3^2$, because J_3 is free, a projective tensored with any module is projective, and projectives over this local algebra are free.

So $V^{\otimes m} = a_m J_1 \oplus b_m J_2 \oplus c_m J_3$ with $(a_{m+1}, b_{m+1}, c_{m+1}) = (b_m, a_m, b_m + 2c_m)$, starting from $(1, 0, 0)$. Hence $a_m + b_m = 1$, $b_m = [m \text{ odd}]$, and counting dimensions, $c_m = (2^m - 1 - [m \text{ odd}])/3$. This gives $f(m) = 1 + (2^m - 1 - [m \text{ odd}])/3$. Summing $\sum_m \binom{d}{m}(q-2)^{d-m} f(m)$ gives the formula. □

6. The range $p \leq d$: tilting lattices

By Theorem E the only pieces not yet described are $X_q(m, c)$ with $m \geq p$. For $p \leq m$ the strings of length $> p$ break: by (1) the coefficient $i(m - 2k - i + 1)$ vanishes modulo p. The way they break is governed by the representation theory of SL_2 in characteristic p.

The dictionary. Let $V = \mathbb{Z}_p v_0 \oplus \mathbb{Z}_p v_1$, where $\mathbb{Z}_p$ is the ring of p-adic integers, and let f act on V by $f v_0 = v_1$, $f v_1 = 0$; give v_0 weight 1 and v_1 weight -1. Identify $B_m \otimes \mathbb{Z}_p$ with $V^{\otimes m}$ by $S \mapsto v_{\mathbf{1}_S}$. Then U is the action of f on the tensor power, a subset of size s has weight $m - 2s$, and so $H = (m - h)/2$ with h the weight. The divided powers $U^{(r)}$ (add r new elements, in all ways) are integral, and so is the analogous operator $e^{(r)}$ (remove r elements). Hence $B_m \otimes \mathbb{Z}_p$ is a module over the distribution algebra $\operatorname{Dist}(SL_2)_{\mathbb{Z}_p}$, and

$$q(c + H) + U \;=\; q\big(c + \tfrac{m-h}{2}\big) + f$$

is built from that action alone.

We use three standard facts.

(T1) $V^{\otimes m}$ is a tilting module, and it decomposes over $\mathbb{Z}_p$ as $\bigoplus_\lambda T_{\mathbb{Z}_p}(\lambda)^{n_\lambda}$, where $T_{\mathbb{Z}_p}(\lambda)$ is the unique lift of the indecomposable tilting module $T(\lambda)$ over $\mathbb{F}_p$. The reason is that $\operatorname{Ext}^1(\Delta, \nabla) = 0$, so decompositions over $\mathbb{F}_p$ lift; this is standard [Jan03, Don93]. For $p = 2$ the argument is written out, and checked in Lean, in [DST, Lemma 3.3, Corollary 3.4].

(T2) For $\lambda \leq p - 1$, and for $\lambda + 1 \equiv 0 \pmod p$ with $\lambda \leq p^2 - 1$, $T(\lambda) = \Delta(\lambda)$ is simple. For $\lambda = p - 1 + r + pa$ with $1 \leq r \leq p - 1$ and $0 \leq a \leq p - 1$, $T(\lambda) \cong T(p - 1 + r) \otimes \Delta(a)^{[1]}$ [Don93], and its Weyl factors are $\Delta(\lambda)$ and $\Delta(\lambda^-)$, $\lambda^- := \lambda - 2r$ (the other terms of the product of characters cancel in pairs). $T(\lambda)$ is a non-split extension $0 \to \Delta(\lambda) \to T(\lambda) \to \Delta(\lambda^-) \to 0$, and so is its lift over $\mathbb{Z}_p$.

(T3) Comparing characters, the multiplicities in $V^{\otimes m}$ are given, from the top down, by $n_\lambda = \beta(m, \lambda) - n_{\lambda^+}$. Here λ^+ is the weight whose bottom factor is λ, and $n_{\lambda^+} = 0$ if there is none or if $\lambda^+ > m$.

For $\lambda \equiv m \pmod 2$, $0 \leq \lambda \leq m$, write $\beta(m, \lambda) = \binom{m}{(m-\lambda)/2} - \binom{m}{(m-\lambda)/2-1}$ (the number of strings of length $\lambda + 1$) and

$$A_c(m, \lambda) = \sum_{r=(m-\lambda)/2}^{(m+\lambda)/2} \big(e + v_p(c + r)\big),$$

the valuation of the determinant of $q(c + H) + U$ on one such string.

The blocks. For $\lambda \equiv m \pmod 2$ define the abelian p-group $G_c(m,\lambda)$ by

$$G_c(m,\lambda) = \begin{cases} \mathbb{Z}/p^{A_c(m,\lambda)}, & \lambda \le p-1, \\ (\mathbb{Z}/p)^a \oplus \mathbb{Z}/p^{A_c(m,\lambda)-a}, & \lambda+1 = (a+1)p,\ a \ge 1, \\ (\mathbb{Z}/p)^{2a} \oplus \mathbb{Z}/p^{A_c(m,\lambda^-)+1-a} \oplus \mathbb{Z}/p^{A_c(m,\lambda)-1-a}, & \lambda = p-1+r+pa, \\ & 1 \le r \le p-1, \end{cases}$$

with the convention that when $c=0$ and $\lambda = m$ (the block containing the empty set) the last factor is replaced by the free summand $\mathbb{Z}_p$.

The first theorem is the statement conjectured in version 1 of this paper. It is the case $m \le 2p-2$ of Theorem G below, where only $a=0$ occurs.

Theorem F (the Glue Theorem; Conjecture F of version 1)**.** *Let p be a prime, q an integer with $e = v_p(q) \ge 1$, $m \le 2p-2$ and $c \ge 0$. Then*

$$\begin{aligned} \operatorname{Syl}_p X_q(m,c) \;\cong\; & \bigoplus_{\lambda \le p-1} \big(\mathbb{Z}/p^{A_c(m,\lambda)}\big)^{\beta(m,\lambda)-\beta'(m,\lambda)} \\ & \oplus \bigoplus_{p \le \lambda \le m} \Big(\mathbb{Z}/p^{A_c(m,\lambda)-1} \oplus \mathbb{Z}/p^{A_c(m,2p-2-\lambda)+1}\Big)^{\beta(m,\lambda)}, \end{aligned}$$

where $\beta'(m,\lambda) = \beta(m,2p-2-\lambda)$ if $\lambda \le p-2$ and $\beta'(m,p-1) = 0$, and λ runs over integers $\equiv m \pmod 2$. When $c=0$ the string of length $m+1$ (through the empty set) has a zero on its diagonal: for $\lambda = m < p$ it gives nothing, and for $\lambda = m \ge p$ the term $\mathbb{Z}/p^{A_c(m,m)-1}$ is omitted (it is the free summand) while its partner $\mathbb{Z}/p^{A_0(m,2p-2-m)+1}$ stays.

More generally:

Theorem G (the tilting blocks)**.** *Let p be a prime, q an integer with $e = v_p(q) \ge 1$, $c \ge 0$, and $m \le 3p-1$ if p is odd, $m \le 3$ if $p = 2$. Then*

$$\mathbb{Z}_p^{[c=0]} \oplus \operatorname{Syl}_p X_q(m,c) \;\cong\; \bigoplus_{\lambda} G_c(m,\lambda)^{\,n_\lambda},$$

where λ runs over the integers $\equiv m \pmod 2$ with $0 \le \lambda \le m$ and n_λ is as in (T3).

With Theorem A, Theorem G determines $\operatorname{Syl}_p K(H(d,q))$ for every $d \le 3p-1$ (p odd). For $p=2$ the range is $d \le 3$, and the whole 2-part of the hypercube is [DST]; for $p=2$ and $m=4$ the formula fails, because the block $\lambda = 4$ already has $a = p-1$ (checked). The hypothesis $e \ge 1$ is needed: for $p \nmid q$ the formula fails, for instance when $(q,p,d) = (4,3,4)$; there the answer of [JNR03] applies instead.

Conjecture H. *For every odd prime p, the isomorphism of Theorem G holds for every $m \le p^2 - 2$.*

The bound is where the method of the proof stops. In Lemma 7 below the denominator of the glue is controlled by the coefficient $(a+1)p$ of e, and for $a = p-1$ this becomes p^2. Indeed for $p=3$ and $m=9$ the formula of Theorem G, applied beyond its range, is wrong by exactly one factor, in the block $\lambda = 9$ (Section 8).

Proofs. Fix $\lambda = p-1+r+pa$ with $1 \le r \le p-1$, $a \in \{0,1\}$ and $a+1 < p$, and put $\mu = \lambda^- = \lambda - 2r$. Let $\ell_i = f^{(i)}x$ $(0 \le i \le \lambda)$ be the standard basis of $\Delta_{\mathbb{Z}_p}(\lambda)$, so $f\ell_i = (i+1)\ell_{i+1}$ and $e\ell_i = (\lambda - i + 1)\ell_{i-1}$. Let $s_j = f^{(j)}y$ $(0 \le j \le \mu)$ be that of $\Delta_{\mathbb{Z}_p}(\mu)$.

Lemma 7 (the glue is forced)**.** *$T_{\mathbb{Z}_p}(\lambda)$ is isomorphic to the lattice in $\Delta_{\mathbb{Q}_p}(\lambda) \oplus \Delta_{\mathbb{Q}_p}(\mu)$ spanned by the ℓ_i and by*

$$\tilde{s}_j = s_j + \tfrac{1}{p}\tbinom{j+r}{j}\,\ell_{j+r} \qquad (0 \le j \le \mu).$$

Proof. By (T2), $T = T_{\mathbb{Z}_p}(\lambda)$ contains $\Delta_{\mathbb{Z}_p}(\lambda)$ with quotient $\Delta_{\mathbb{Z}_p}(\mu)$. Weight spaces of $\Delta(\lambda)$ are one-dimensional, so the lifts of the s_j to $T \subset \Delta_{\mathbb{Q}_p}(\lambda) \oplus \Delta_{\mathbb{Q}_p}(\mu)$ have the form $\tilde{s}_j = s_j + x_j \ell_{j+r}$, $x_j \in \mathbb{Q}_p$, determined modulo $\mathbb{Z}_p$.

Stability under e at $j = 0$ gives $x_0(\lambda - r + 1) \in \mathbb{Z}_p$, and $\lambda - r + 1 = (a+1)p$ with $a + 1 < p$, so $x_0 \in p^{-1}\mathbb{Z}_p$.

Stability under f gives $(j+r+1)x_j - (j+1)x_{j+1} \in \mathbb{Z}_p$. For $j \not\equiv -1 \pmod p$ the factor $j+1$ is a unit, so $x_{j+1} \equiv \frac{j+r+1}{j+1} x_j$ modulo $\mathbb{Z}_p$, and within each block of indices $[kp, (k+1)p-1]$ we get $x_j \equiv \binom{j+r}{j}\binom{kp+r}{kp}^{-1} x_{kp}$. At $j = p-1$ the condition reads $r x_{p-1} \in \mathbb{Z}_p$, and it holds automatically: by Lucas, $p \mid \binom{p-1+r}{p-1}$. When $a = 1$, stability under $f^{(p)}$ at $j = 0$ gives $\binom{r+p}{p} x_0 - x_p \in \mathbb{Z}_p$, and $\binom{r+p}{p} \equiv 1$ by Lucas, so $x_p \equiv x_0$. Hence $x_j \equiv \binom{j+r}{j} x_0$ for every j.

If $x_0 \in \mathbb{Z}_p$ then $T = \Delta_{\mathbb{Z}_p}(\lambda) \oplus \Delta_{\mathbb{Z}_p}(\mu)$, which is impossible since $T \otimes \mathbb{F}_p$ is indecomposable. So $x_0 = u/p$ with u a unit, and rescaling y makes $u = 1$. The lattice so described is T itself, so it is stable under all $f^{(t)}$ and $e^{(t)}$; nothing more has to be checked. (For $f^{(t)}$ this is also visible directly: $f^{(t)}\tilde{s}_j - \binom{j+t}{t}\tilde{s}_{j+t}$ vanishes by the identity $\binom{j+r}{j}\binom{j+r+t}{t} = \binom{j+t}{t}\binom{j+t+r}{j+t}$.) Stability under every $e^{(t)}$ and $f^{(t)}$ is also checked by machine in Section 7. □

Lemma 8 (the blocks)**.** *Embed $T_{\mathbb{Z}_p}(\lambda)$ in $B_m \otimes \mathbb{Z}_p$ at levels $k, \dots, k+\lambda$ ($m = \lambda + 2k$). Then the cokernel of $q(c+H)+U$ on it is $G_c(m, \lambda)$. The same holds for the simple blocks ($\lambda \le p-1$, or $\lambda + 1 \in p\mathbb{Z}$).*

Proof. Put $N_t = q(c + k + t)$, so $v_p(N_t) = e + v_p(c + k + t)$, and $M = q(c + H) + U$. In the basis of Lemma 7:

- $M\ell_i = N_i \ell_i + (i+1)\ell_{i+1}$;
- $M\tilde{s}_j = N_{j+r}\tilde{s}_j + (j+1)\tilde{s}_{j+1}$ for $j < \mu$ (the ℓ-component is $\frac{1}{p}[(j+r+1)\binom{j+r}{j} - (j+1)\binom{j+r+1}{j+1}]\ell_{j+r+1} = 0$);
- $M\tilde{s}_\mu = N_{\mu+r}\tilde{s}_\mu + u\,\ell_{\mu+r+1}$, where $u = \frac{1}{p}\binom{\mu+r}{\mu}(\mu + r + 1) = (a+1)\binom{(a+1)p-1}{\mu}$ is a unit by Lucas.

The non-unit subdiagonal entries are the $i + 1 \in \{p, \dots, (a+1)p\}$ on the ℓ-chain and the $j + 1 \in \{p, \dots, ap\}$ on the $\tilde{s}$-chain, each of valuation 1. Solving the unit subdiagonals, the cokernel is generated by the first vector of each segment between them, with one relation at the end of each segment.

For $a = 0$ the generators are $\ell_0, \ell_p, \tilde{s}_0$. With $\alpha = \sum_{t<p} v(N_t)$, $\beta_2 = \sum_{t \ge p} v(N_t)$ and $A' = \sum_{t=r}^{p-1} v(N_t) = A_c(m, \mu)$, the relations become (up to units) $p^\alpha \ell_0 + p\,\ell_p$, $p^{\beta_2}\ell_p$, and $\ell_p = p^{A'}\tilde{s}_0$. So the cokernel is presented on $(\ell_0, \tilde{s}_0)$ by $\begin{pmatrix} p^\alpha & p^{1+A'} \\ 0 & p^{\beta_2 + A'} \end{pmatrix}$. Since $\alpha = A' + \sum_{t<r} v(N_t) \ge A' + 1$ and $\beta_2 \ge 1$, its elementary divisors are $p^{A'+1}$ and $p^{\alpha+\beta_2-1} = p^{A-1}$, with $A = A_c(m, \lambda)$.

For $a = 1$ the generators are $\ell_0, \ell_p, \ell_{2p}, \tilde{s}_0, \tilde{s}_p$, and ℓ_{2p} is eliminated by the cross relation. With segment valuations a_1, a_2, a_3 on the ℓ-chain and σ_1, σ_2 on the $\tilde{s}$-chain, the presentation on $(\ell_0, \ell_p, \tilde{s}_0, \tilde{s}_p)$ is

$$\begin{pmatrix} p^{a_1} & p & 0 & 0 \\ 0 & p^{a_2} & 0 & p^{1+\sigma_2} \\ 0 & 0 & p^{\sigma_1} & p \\ 0 & 0 & 0 & p^{a_3+\sigma_2} \end{pmatrix}.$$

Every entry has valuation ≥ 1 and p occurs twice in different rows and columns, so the first two elementary divisors are p. The smallest valuation of a 3×3 minor is $2 + \sigma_1 + \sigma_2 = 2 + A'$, because $a_1 + a_2 \ge A' + 1$ and $a_3 \ge 1$. The determinant has valuation $A + A'$. So the divisors are $p, p, p^{A'}, p^{A-2}$, which is $G_c(m, \lambda)$ for $a = 1$.

When $c = 0$ and $k = 0$, $N_0 = 0$: the vector ℓ_0 generates a free summand, and the relation at the end of the first segment loses its first term. The same elimination gives $\mathbb{Z}_p \oplus$ the stated group. The simple blocks are single chains; when $\lambda + 1 = (a+1)p$ the chain has a non-unit subdiagonal entries, each p, and the same elimination gives $(\mathbb{Z}/p)^a \oplus \mathbb{Z}/p^{A-a}$. □

Proof of Theorems G and F. Combine (T1)–(T3), Lemma 7 and Lemma 8. In the stated range every $\lambda \le m$ is simple or has $a \le 1$ and $a+1 < p$. □

Example 9. For $p = q = 3$ and $d = 4$: Theorem A gives the pieces $X(4,0)$, $X(3,1)^4$, $X(2,2)^6$, $X(1,3)^4$, $X(0,4)$, and Theorem F gives $\mathrm{Syl}_3 K(H(4,3)) \cong (\mathbb{Z}/3)^2 \oplus (\mathbb{Z}/3^2)^7 \oplus (\mathbb{Z}/3^3)^8 \oplus (\mathbb{Z}/3^4)^{17}$, which is the group we computed.

7. Verification

All computations were exact. The Smith forms over $\mathbb{Z}_{(p)}$ were computed modulo p^M with $p^M < 2^{31}$, by unit pivots and Schur complements, level by level in the p-adic valuation; factors of valuation at least M were then determined separately, by the matrix $p^E M^{-1}$ of Lemma 4, whose elementary divisors are p^{E-a} for the divisors p^a of M. The scripts, the sealed prediction files and the logs are at `https://github.com/tretoef-estrella/coloured-sand-theorem` and in the Zenodo record doi:10.5281/zenodo.23327608 (all versions).

(1) Theorem A: the Sylow p-subgroups for $p \in \{2,3,5,7\}$ computed directly from the Laplacian and from the pieces agree in all 90 comparisons (27 cells, $q \in \{3,\dots,10,12\}$, $q^d \le 729$, including $p \mid q$ and $p \le d$); adding one extra copy of a piece changes the answer in 49 of them. Independently, the cells $(d,q) = (3,7)$ and $(2,11)$ were recomputed with Sage from the full Laplacian.

(2) Theorem B: found first as a conjecture from the cells with $d \le 4$; before the proof, ten predictions for $(d,p) = (2,11), (2,13), (2,17), (2,19), (2,23), (3,17), (3,19), (3,23), (3,29), (4,13)$ were written to a file (MD5 `d5c36be3736079cbc2349f469751c627`) and then computed: ten agreements.

(3) Theorem C: 143 cells, for $q \in \{3,\dots,10,12\}$ and $p \in \{2,3,5,7\}$, both with $p \mid q$ (including $p = 2$) and with $p \nmid q$, all in agreement; the guess $ed + v_p(d)$ fails in one of them and the guess ed in 26.

(4) Theorem D: the rank formula for $W_{s,s+1}$ checked directly in 264 cases ($p \le 7$, $m \le 11$); the value of N agrees with the computed groups in 23 cells with $p \le d$.

(5) Theorem F, before its proof (version 1): found from 88 pieces with $p = 3, 5$. Then 30 pieces with $p = 7$, $7 \le m \le 11$, $1 \le c \le 6$ were predicted before being computed (MD5 of the file `1f5cc12e88f12e375022658da3eea6ce`): 30 agreements. Then 14 full groups $\mathrm{Syl}_p K(H(d,q))$ with composite $q \in \{6,9,10,12,15,20,25,27\}$ and $p \le d \le 2p-2$ (MD5 `341a2c489aef1c30349bae079ddcce81`): 14 agreements. It also agrees with the 10 full groups computed for $q = p \in \{3,5,7\}$ and $p \le d \le 2p-2$.

(6) Lemmas 7 and 8, by machine:
- the lattices of Lemma 7 are stable under every $e^{(t)}$ and $f^{(t)}$ for $p \in \{3,5,7\}$ and every $\lambda \in [p, 3p-2]$ (27 lattices; 2200 checks for $\lambda \le 2p-2$);
- the cokernel of $q(c+H)+U$ on the model equals the formula of Lemma 8 in 292 random cases with $\lambda \le 2p-2$ ($p \le 7$), in 180 cases with $\lambda \le p^2-2$ ($p = 3,5$; $e \in \{1,2\}$), and in 60 cases with $c = 0$;
- removing the glue (the split lattice) changes the answer in 293 of 293 cases, and the formula of Theorem F applied beyond $2p-2$ fails in 12 of 18 pieces.

(7) Theorem G against direct computation: the predicted pieces equal the Smith forms of $q(c+H)+U$ computed on B_m in 154 cases ($p = 3$, $m \le 7$; $p = 5$, $m \le 10$; $e \in \{1,2\}$; $0 \le c \le 6$) and in 5 cases with $p = 3$, $m = 8 = 3p-1$. The full groups for $p = q = 3$ and $d = 5,\dots,8$ computed in version 1 from the full Laplacian (Section 8) are reproduced exactly.

(8) Corollary 6: $\dim_{\mathbb{F}_p} \operatorname{coker} U$ on B_m computed by ranks agrees with the count of Jordan blocks in 37 cases ($p \in \{3,5,7\}$, $m \le 12$); the characteristic-0 count $\binom{m}{\lfloor m/2 \rfloor}$ differs in 7 of them.

8. Data for $p = q = 3$

For $q = p = 3$ the computed groups, written as $(n_1, n_2, \dots)$ with n_v the number of factors $\mathbb{Z}/3^v$, are:

d	$(n_1, n_2, \dots)$
5	$(12, 10, 16, 53, 5)$
6	$(38, 37, 36, 117, 24, 16, 6)$
7	$(86, 105, 127, 285, 85, 63, 41)$
8	$(250, 211, 286, 1078, 174, 133, 174, 8)$
9	$(942, 412, 624, 3339, 449, 292, 495, 133, 84, 37, 9)$

Theorem G gives the rows $d = 5, \dots, 8$ exactly ($d \le 3p - 1$). For $d = 9$ one piece has $m = 9 = 3p$, outside the range. There the formula of Theorem G, applied anyway, gives $(942, 414, 623, 3339, 448, 293, 495, 134, 84, 36, 9)$ instead of the true row. The difference is one factor, in the block $\lambda = 9 = 3p$, where $a = p-1$ and the glue of Lemma 7 may have denominator p^2. For $p = 2$ the whole problem is the hypercube, solved in [DST] with the tilting modules of SL_2 in characteristic 2; the general case $m \ge 3p$ should follow the same pattern, with tilting modules having more Weyl factors.

Provenance

The results were found on the night of 12–13 October 2026 by an AI system, Claude (Anthropic), working for the author: it chose the problem after a search of the literature, wrote the computations, found the formulas, sealed and checked the predictions, and found the proofs; it also found, when checking the literature before publication, that Theorem A is [JNR03, Theorem 13]. Version 2 was written later the same night, also by Claude: the Glue Conjecture of version 1 was proved and extended (Theorems F, G) by writing down the tilting lattices, and the closed form of N (Corollary 6) was proved. The author takes responsibility for the paper. To the best of our knowledge, after a search of the works citing [JNR03] and of arXiv, Theorems B, C (for $e \ge 1$), D, E, F and G, Corollary 6 and Conjecture H are new; version 1 also claimed the case $e = 0$ of Theorem C, which follows from [JNR03, Theorem 2]. Theorem A is due to Jacobson, Niedermaier and Reiner. The tilting theory used in Section 6 is classical [Don93, Jan03], and so is the decomposition of tensor powers in the proof of Corollary 6.

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